

Analytical and Numerical Investigation of the Neutron Diffusion and Wave Equations

Louis Miller | 40298094 | Project 2

November 22, 2025

1 INTRODUCTION

This paper investigates the neutron diffusion equation within a spatial domain of fixed length L . Both analytical and numerical methods were employed to determine the critical length, at which the system becomes unstable for Dirichlet boundary conditions, Neumann boundary conditions and a mixture of the two. Additionally, the effects of domain scaling on the critical length were analysed. The second section of the paper focused on the wave equation, for which both numerical and analytical solutions were studied. A comparison is made between these solutions, and the stability of the analytical solution is discussed. Furthermore, the Crank-Nicolson method is used to solve the system, and the results are compared with those obtained from the explicit central finite difference method.

2 NEUTRON DIFFUSION EQUATION

The neutron diffusion equation investigated in this paper was a one-dimension analogue of the fast neutron chain reaction, famously studied in 1943 during the Manhattan Project. The Partial Differential Equation (PDE) took the form:

$$\frac{\partial n(x, t)}{\partial t} = D \frac{\partial^2 n(x, t)}{\partial x^2} + C n(x, t) \quad (2.1)$$

where D and C are the diffusion constant, and creation rate for neutrons respectively. The spatial domain was confined to $-\frac{L}{2} \leq x \leq \frac{L}{2}$ where L denotes the length of the system. Time t progressed starting from $t = 0$, with $t > 0$. The system was set up with Dirichlet boundary conditions, which were:

$$n(x = -\frac{L}{2}, t) = n(x = \frac{L}{2}, t) = 0 \quad (2.2)$$

the physical interpretation of these boundary conditions (2.2) is that no neutrons can exist outside the spatial domain. Finally, the system also included a numerical initial condition:

$$n(x = 0, t = 0) = \frac{1}{\Delta x} \quad \text{for } C = D = 1 \quad (2.3)$$

where Δx is the spatial step size in the numerical solution.

2.1 ANALYTICAL SOLUTION

A solution to equation 2.1 was sought using the method of variable separation:

$$\frac{1}{T(t)} \frac{dT(t)}{dt} = D \frac{d^2 X(x)}{dx^2} + C \quad \text{where} \quad n(x, t) = X(x)T(t) \quad (2.4)$$

since both the right-hand side (RHS) and left hand side (LHS) of equation 2.4 depend solely on different variables, t and x respectively, they must be equal to a constant $-\lambda$. This results in a spatial (x) Ordinary Differential Equation (ODE) and temporal (t) ODE.

$$\frac{1}{T} \frac{dT}{dt} = -\lambda \quad (2.5)$$

$$\frac{D}{X} \frac{d^2 X}{dx^2} + C = -\lambda \quad (2.6)$$

The separation constant was chosen as $-\lambda$ with $\lambda > 0$, such that equation 2.6 results in nontrivial sinusoidal eigen-functions that satisfied the Dirichlet boundary conditions 2.2. Had the opposite sign been chosen, solutions to the spatial ODE 2.6 would not satisfy the Dirichlet boundary conditions (2.2).¹ Solving both ODEs (2.5 & 2.6) yields:

$$T(t) = Ae^{-\lambda t} \quad (2.7)$$

$$X(x) = B \cos(kx) + C \sin(kx) \quad (2.8)$$

$$\text{where } k^2 = \frac{\lambda + C}{D} \quad (2.9)$$

where A , B , and C are all arbitrary constants. By multiplying equations 2.7 & 2.8 a solution to equation 2.1 in terms of these arbitrary constants was obtained. By applying the boundary conditions 2.2 and excluding the trivial solution $A = 0$, the following equations are obtained:

$$B \cos\left(\frac{kL}{2}\right) + C \sin\left(\frac{kL}{2}\right) \quad (2.10)$$

$$B \cos\left(\frac{kL}{2}\right) - C \sin\left(\frac{kL}{2}\right) \quad (2.11)$$

two distinct branches of solutions were formed, depending on how equations 2.10 & 2.11 were treated. Subtracting equation 2.11 from 2.10 results in:

$$2C \sin\left(\frac{kL}{2}\right) = 0 \quad (2.12)$$

assuming $C \neq 0$, equation 2.12 is satisfied when $k = \frac{2\pi n}{L}$ for $n = 1, 2, 3, \dots$. This expression is denoted as k_n . Adding equation 2.10 and 2.11 results in the following expression:

$$2B \cos\left(\frac{kL}{2}\right) = 0 \quad (2.13)$$

applying the same logic as the sine branch, equation 2.13 is satisfied when $k_m = \frac{(2m+1)\pi}{L}$ for $m = 0, 1, 2, 3, \dots$. The following solution to equation 2.1 is found by combining both sine and cosine branches:

$$n(x, t) = \sum_{n=1}^{\infty} A_n e^{-\lambda_n t} \sin(k_n x) + \sum_{m=1}^{\infty} B_m e^{-\lambda_m t} \cos(k_m x) \quad (2.14)$$

where $\lambda_n = Dk_n^2 - C$ and $\lambda_m = Dk_m^2 - C$

¹The general solution to 2.6 with an opposite sign for λ is: $X(x) = Ae^{\kappa x} + Be^{-\kappa x}$ where $\kappa^2 = \frac{\lambda - C}{D}$, the only solution that satisfies the boundary conditions 2.2 is the trivial solution, where $A = B = 0$.

To simplify the analysis, a change of variables was introduced: $y = x + \frac{L}{2}$. This transformation does not alter the structure of the PDE in equation 2.1,² but does restrict the spatial domain to $0 \leq y \leq L$ given the original domain $-\frac{L}{2} \leq x \leq \frac{L}{2}$.

$$n(y = 0, t) = 0 \quad (2.15)$$

$$n(y = L, t) = 0 \quad (2.16)$$

Since the structure of equation 2.1 remains unchanged except for the variable substitution, the general form of equation 2.8 is preserved, with x replaced by y and by applying the boundary condition 2.15, $B = 0$ is obtained, thereby reducing the spatial part of the equation to a sine term: $C \sin(ky)$. Finally, applying the boundary condition 2.16 and substituting $y = x + \frac{L}{2}$ yields a solution with the unknown coefficient A_n :

$$n(x, t) = \sum_{n=1}^{\infty} A_n \exp \left(\underbrace{\left(C - \frac{Dn^2\pi^2}{L^2} \right)}_{-\lambda_n} t \right) \sin \left(\frac{n\pi(x + L/2)}{L} \right) \quad (2.17)$$

from equation 2.17, the minimum length after which neutron density always increases as a function of time (the critical length) was found by identifying when the exponential term in equation 2.17 was positive, that was, when $C - \frac{Dn^2\pi^2}{L^2} = 0$ for $n = 1$. An important observation is that for $n = 0$ the sine term in equation 2.17 reduces to $\sin(0)$ resulting in $n(x, t) = 0$. Therefore, the minimum value for n is one. This value of n was used to calculate the critical length in this case.

$$L_{critical} = \pi \sqrt{\frac{D}{C}} \quad (2.18)$$

To find A_n in equation 2.17, C and D were set equal to one and the following initial condition was used:

$$n(x = 0, t = 0) = \delta(x) \quad (2.19)$$

where $\delta(x)$ is the Dirac delta function. The initial condition 2.19 is an analytical approximation to the numerical initial condition (2.3). However, it is useful to have a complete analytical solution to compare three-dimensional graphs plotted by both methods. By applying the initial conditions and using the orthogonality of sine functions under integration, the coefficient A_n was determined:

$$\begin{aligned} n(x, 0) &= \sum_{n=1}^{\infty} A_n \sin \left(\frac{n\pi(x + L/2)}{L} \right) \\ \int_{-\frac{L}{2}}^{\frac{L}{2}} n(x, 0) \sin \left(\frac{m\pi(x + L/2)}{L} \right) dx &= \frac{A_n L}{2} \delta_{mn} \\ \frac{2}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta(x) \sin \left(\frac{n\pi(x + L/2)}{L} \right) dx &= A_n \\ \frac{2}{L} \sin \left(\frac{n\pi}{2} \right) &= A_n \end{aligned} \quad (2.20)$$

where δ_{mn} denotes the Kronecker delta. By combining the coefficient A_n from equation 2.20 with the solution in equation 2.17 a complete analytical solution to equation 2.1 was obtained.

$$n(x, t) = \frac{2}{L} \sum_{n=1}^{\infty} \sin \left(\frac{n\pi}{2} \right) \exp \left(\left(1 - \frac{n^2\pi^2}{L^2} \right) t \right) \sin \left(\frac{n\pi(x + L/2)}{L} \right) \quad (2.21)$$

The analytical solution was plotted and is shown in Figure 2.1.

²Performing the change of variables $y = x + \frac{L}{2}$ to equation 2.1 yields the PDE: $n_t = Dn_{yy} + Cn$.

2.2 NUMERICAL SOLUTION

In this section, the neutron diffusion equation (2.1) was solved using the finite difference method. Firstly a grid of values of x and t for equation 2.1 was defined:

$$x_i = i\Delta x \quad \text{for } i = 0, 1, 2, \dots, N-1 \quad \text{where } \Delta x = \frac{L}{N-1} \quad (2.22)$$

$$t_j = j\Delta t \quad \text{for } j = 0, 1, 2, \dots \quad (2.23)$$

where Δt is dependent on how many j values there were. To ensure clarity and thoroughness, the derivation of the finite difference equation for equation 2.1 is provided. The notation $n(x_i, t_j) \equiv n_{i,j}$ is used throughout, where $n_{i,j}$ denotes the value of n at the spatial point x_i and time step t_j . By performing a Taylor expansion of $n(x_{i+1}, t_j)$ and $n(x_i, t_{j+1})$ about the point (x_i, t_j) , approximations for the spatial and temporal derivatives were obtained respectively.

$$n(x_{i+1}, t_j) = n(x_i, t_j) + \Delta x \frac{\partial}{\partial x} n(x_i, t_j) + \frac{(\Delta x)^2}{2} \frac{\partial^2}{\partial x^2} n(x_i, t_j) + O((\Delta x)^3) \quad (2.24)$$

$$n(x_i, t_{j+1}) = n(x_i, t_j) + \Delta t \frac{\partial}{\partial t} n(x_i, t_j) + O((\Delta t)^2) \quad (2.25)$$

The two point forward approximation for the first order temporal derivative was found by rearranging equation 2.25.

$$\frac{\partial}{\partial t} n(x_i, t_j) = \frac{n_{i,j+1} - n_{i,j}}{\Delta t} \quad (2.26)$$

Performing a Taylor expansion around $n(x_{i-1}, t_j)$ about the point (x_i, t_j) resulted in:

$$n(x_{i-1}, t_j) = n(x_i, t_j) - \Delta x \frac{\partial}{\partial x} n(x_i, t_j) + \frac{(\Delta x)^2}{2} \frac{\partial^2}{\partial x^2} n(x_i, t_j) + O((\Delta x)^3). \quad (2.27)$$

Adding equation 2.27 and 2.24, a three point central difference approximation for the second order spatial derivative was obtained.

$$\frac{\partial^2}{\partial x^2} n(x_i, t_j) = \frac{n_{i+1,j} - 2n_{i,j} + n_{i-1,j}}{(\Delta x)^2} \quad (2.28)$$

Substituting equations 2.26 and 2.28 into equation 2.1 and rearranging for $n_{i,j+1}$ yielded:

$$n_{i,j+1} = \alpha n_{i-1,j} + \beta n_{i,j} + \alpha n_{i+1,j} \quad (2.29)$$

$$\text{where } \alpha = \frac{D\Delta t}{(\Delta x)^2} \quad \text{and} \quad \beta = (1 + C\Delta t - 2\alpha). \quad (2.30)$$

The computation of $n(x_i, t_{j+1})$ using equation 2.29 is more effectively expressed and implemented with the use of matrices. Let $\mathbf{W}^{(j)}$ denote a vector of solutions to equation 2.1 at time t_j , defined as

$$\mathbf{W}^{(j)} = \begin{bmatrix} n_{1,j} \\ n_{2,j} \\ \vdots \\ n_{N-2,j} \end{bmatrix} \quad (2.31)$$

an important note is that $\mathbf{W}^{(j)}$ excludes $n_{0,j}$ as the first element and $n_{N-1,j}$ as the last element, since these are boundary solutions (2.2) and hence equal to zero.³ Starting from $i = 0$ in equation 2.29 i.e. $i = 0$ is the first spatial gap, leads to a system of linear equations.

$$\begin{aligned} n_{1,j+1} &= \beta n_{1,j} + \alpha n_{2,j} \quad \text{where } \alpha n_{0,j} = 0 \\ n_{2,j+1} &= \alpha n_{1,j} + \beta n_{2,j} + \alpha n_{3,j} \\ n_{3,j+1} &= \alpha n_{2,j} + \beta n_{3,j} + \alpha n_{4,j} \\ &\vdots \\ n_{N-3,j+1} &= \alpha n_{N-4,j} + \beta n_{N-3,j} + \alpha n_{N-2,j} \\ n_{N-2,j+1} &= \alpha n_{N-3,j} + \beta n_{N-2,j} \quad \text{where } \alpha n_{N-1,j} = 0 \end{aligned} \quad (2.32)$$

³The first spatial step at time step j corresponds to $x = -L/2$, the last spatial step is $x = L/2$. Hence, $n_{0,j} = n(x = -L/2, t = j\Delta t) = 0$ and $n_{N-1,j} = n(x = L/2, t = j\Delta t) = 0$. This holds true $\forall j$

The system of linear equations presented in 2.32 can be used to describe a tridiagonal coefficient matrix $\mathbf{B}$, where the zero coefficients from $n_{0,j}$ and $n_{N-1,j}$ are omitted.⁴

$$\mathbf{B} = \begin{bmatrix} \beta & \alpha & & & \\ \alpha & \beta & \alpha & & \\ & \alpha & \beta & \alpha & \\ & & \alpha & \ddots & \ddots \\ & & & \ddots & \ddots & \alpha \\ & & & & \alpha & \beta \end{bmatrix} \quad (2.33)$$

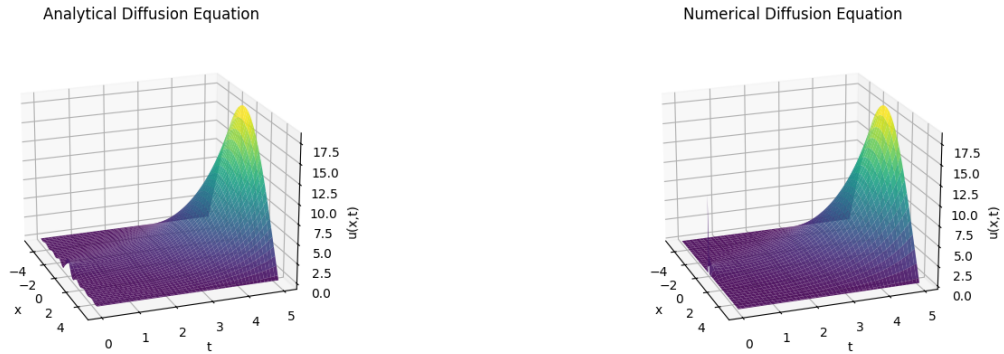
Entries corresponding to empty spaces in the coefficient matrix shown in 2.33 are given as zero, α and β were taken from equation 2.30. Multiplying the matrix 2.33 by the column vector 2.31, the finite difference equation (2.29) was written in a matrix form.

$$\mathbf{W}^{(j+1)} = \mathbf{B}\mathbf{W}^{(j)} \quad (2.34)$$

Code was written in Python using equation 2.34 to compute each time-step solution vector, the algorithm was initiated with a vector evaluated using the numerical initial condition (2.3), this vector took the form:

$$(\mathbf{W}^{(0)})^T = [0 \quad 0 \quad \dots \quad 0 \quad \frac{1}{\Delta x} \quad 0 \quad \dots \quad 0]. \quad (2.35)$$

Vectors produced by equation 2.34 were transposed and constructed row-wise into a solution matrix, with the first row representing time step zero, the next row time step one, and so forth. To enforce boundary conditions 2.2, zero columns were appended to the left and right of the solution matrix. Each row was then plotted against the spatial grid using Matplotlib [1]. Both the analytical and Numerical solution plots produced are shown in Figure 2.1. An important consideration is that



(a) Analytical solution to the neutron diffusion PDE (2.1), using equation 2.21 for $n = 1, 2, \dots, 15$. (b) Numerical solution to the neutron diffusion PDE (2.1), using the finite difference method discussed in section 2.2.

Figure 2.1: Plots (a) and (b) present an analytical and numerical solution to equation 2.1 for $L = 10$ up to time $t = 5$. In both plots, $C = D = 1$ and $u(x, t) \equiv n(x, t)$.

the following condition had to be satisfied to plot the numerical solution:

$$\Delta t < \frac{(\Delta x)^2}{2}. \quad (2.36)$$

To approximate a critical length numerically, the root-mean-square (RMS) of each row in the solution space corresponding to a single time step was computed. This quantity provided a scalar measure

⁴This is a consequence of removing boundary solutions from 2.31

of the amplitude of the solution at each time step. This amplitude is denoted by $A(t)$. This provided a vector comprised of amplitudes at each time step in the system.

$$\mathbf{A}_t = [A(0), A(\Delta t), \dots, A(M\Delta t)] \quad (2.37)$$

Where M in equation 2.37 is the total number of time steps. Both numerical observation (see Figure 2.1) and analytical analysis of the neutron diffusion equation (Section 2.1) indicate that the solution exhibits exponential behaviour (equation 2.17) of the form:

$$A(t) = A_0 e^{\lambda_A t}, \quad (2.38)$$

for some growth rate λ_A , this leads to the following linear relation:

$$\ln(A(t)) = \lambda_A t + C \quad (2.39)$$

allowing λ_A to be estimated using a least-squares fit on equation 2.39. This procedure was applied for a range of L values. The critical length was then identified as the value of L for which λ_A satisfied $\lambda_A \approx 0$, the approximation carried out by this algorithm is seen in Figure 2.2. The numerical critical

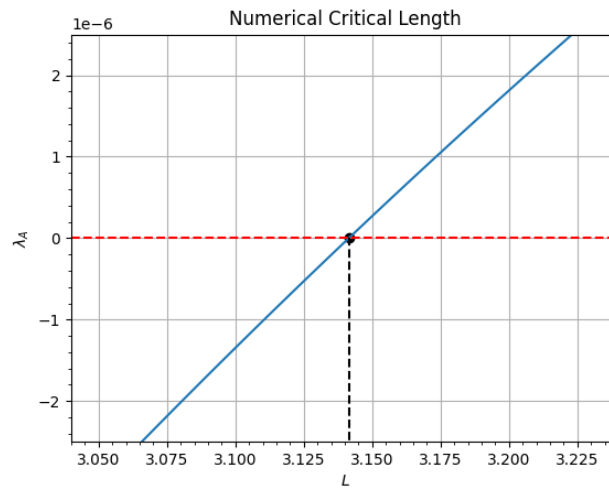


Figure 2.2: An approximation for the critical length was found by an exponential fit described in Section 2.2. λ_A is the gradient to the approximating exponential curve (2.39). The red dashed lined is a visual indication for the x axis, or in this case the λ_A axis. The dashed black line indicates where the approximate critical length $L_{critical}$ is. From the figure, a value for the critical length is:
 $L_{critical} = 3.142$ to three decimal places.

length calculated was $L_{critical} = 3.142$ to three decimal places. This closely matches the analytical critical length seen in equation 2.18 when $C = D = 1$.

The graphs produced by the analytical and numerical solutions (Figure 2.1) exhibit strong agreement, with noticeable discrepancies within a small region close to the initial time step. This is due to the analytical initial condition (2.19) being a representation of the numerical initial condition (2.3). Although not exact, both figures exhibit the same dependence on time and critical length analysis. Finally, an animation for both figures (2.1), where the values of L are varied is available on GitHub [2].

2.3 NEUMANN & DIRICHLET BOUNDARY CONDITIONS

In this section, the boundary conditions for the neutron diffusion PDE 2.1 were modified to Neumann boundary conditions, which were:

$$\frac{\partial n(x = -\frac{L}{2}, t)}{\partial x} = 0 \quad (2.40)$$

$$\frac{\partial n(x = \frac{L}{2}, t)}{\partial x} = 0. \quad (2.41)$$

The boundary conditions were altered to examine the critical length further and to determine the impact that the boundary conditions had on $L_{critical}$. Initially, a single boundary condition was changed to a Neumann boundary condition. The subsequent analysis will contain equation 2.40 alongside $n(x = \frac{L}{2}, t) = 0$ from equation 2.2 as the boundary conditions. Beginning with the analytical approach, the analysis proceeded with the use of equations 2.7 and 2.8. The same change of variables that was made in section 2.1 was also made here: $y = x + \frac{L}{2}$, this resulted in the following two conditions:

$$\frac{\partial n(y = 0, t)}{\partial y} = 0 \quad (2.42)$$

$$n(y = L, t) = 0 \quad (2.43)$$

applying condition 2.42, the subsequent equation was found:

$$\frac{\partial n(y = 0, t)}{\partial y} = Ck \cos(0) - Bk \sin(0) = 0 \quad (2.44)$$

$$\text{therefore } C = 0$$

lastly, applying condition 2.43 results in:

$$B \cos(kL) = 0 \quad (2.45)$$

similar to section 2.1, equation 2.45 is satisfied when $k = \frac{(2m+1)\pi}{2L}$ for $m = 0, 1, 2, \dots$. From this point, the critical length can be determined. An important note of comparison is the allowed minimum value for m , which is zero, as well as the form of k , which arises when the cosine function equals zero. The neutron diffusion PDE (2.1) with Dirichlet boundary conditions (2.2) had allowed k values corresponding to when sine equaled zero and a minimum value of $n = 1$. The resultant critical length was:⁵

$$L_{critical} = \frac{\pi}{2} \sqrt{\frac{D}{C}}. \quad (2.46)$$

The critical length calculated in equation 2.46 is half of the critical length determined in equation 2.18, which was derived for the system using only Dirichlet boundary conditions (2.2). Using similar reasoning when applying the analytical initial condition (2.19) in section 2.1, a complete analytical solution to equation 2.1 for boundary conditions 2.40 and $n(x = \frac{L}{2}, t) = 0$ from equation 2.2 was found.

$$n(x, t) = \frac{2}{L} \sum_{m=0}^{\infty} \cos\left(\frac{(2m+1)\pi}{4}\right) \cos\left(\frac{(2m+1)\pi}{2L}\left(x + \frac{L}{2}\right)\right) \exp\left(\left(1 - \left(\frac{(2m+1)\pi}{2L}\right)^2\right)t\right) \quad (2.47)$$

A numerical approach was applied for the same boundary conditions (Neumann condition in equation 2.40 and $n(x = \frac{L}{2}, t) = 0$ from equation 2.2). The three-point central difference approximation for a first order partial derivative was derived by subtracting equation 2.27 from 2.24.

$$\frac{\partial}{\partial x} n(x_i, t_j) = \frac{n_{i+1,j} - n_{i-1,j}}{2\Delta x} \quad (2.48)$$

By using the spatial index corresponding to $i - 1$ as a ghost point [3] since it lies outside of the spatial domain ($-\frac{L}{2} \leq x \leq \frac{L}{2}$), and applying the boundary condition 2.40 to equation 2.48, the following result was obtained:

$$n_{2,j} = n_{0,j} \quad (2.49)$$

this caused the coefficient matrix governing this system to differ from the one seen in equation 2.33.

$$\mathbf{B}_2 = \begin{bmatrix} \beta & 2\alpha & & & & \\ \alpha & \beta & \alpha & & & \\ & \alpha & \beta & \alpha & & \\ & & \alpha & \ddots & \ddots & \\ & & & \ddots & \ddots & \alpha \\ & & & & \alpha & \beta \end{bmatrix} \quad (2.50)$$

⁵The critical length is determined by rearranging: $C - D\left(\frac{(2m+1)\pi}{2L}\right)^2 = 0$ to find L , for $m = 0$

An important distinction to make is that, similar to the coefficient matrix (2.33) yielded from the Dirichlet boundary conditions (2.2), the matrix 2.50 contains coefficients that exclude the boundary conditions. However, from equation 2.50 the resulting solution matrix only had a single column of zeros appended at the end, rather than the beginning as well. This was a consequence of changing the boundary condition corresponding to $x = -L/2$. Another important detail is that, instead of using equation 2.48, the two-point forward difference approximation could be applied, resulting in:

$$\frac{\partial}{\partial x} n(x_i, t_j) = \frac{n_{i+1,j} - n_{i,j}}{\Delta x} \quad (2.51)$$

and applying the same boundary condition 2.40 to equation 2.51 yielded:

$$n_{1,j} = n_{0,j} \quad (2.52)$$

this gave two different matrices depending on which index was replaced.

$$\mathbf{B}_2^{(1)} = \begin{bmatrix} (\beta + \alpha) & 0 & \alpha & & & \\ \alpha & \beta & \alpha & & & \\ & \alpha & \beta & \alpha & & \\ & & \alpha & \ddots & \ddots & \\ & & & \ddots & \ddots & \alpha \\ & & & & \ddots & \ddots & \alpha & \beta \end{bmatrix} \quad \text{or} \quad \mathbf{B}_2^{(2)} = \begin{bmatrix} 0 & (\beta + \alpha) & \alpha & & & \\ \alpha & \beta & \alpha & & & \\ & \alpha & \beta & \alpha & & \\ & & \alpha & \ddots & \ddots & \\ & & & \ddots & \ddots & \alpha \\ & & & & \ddots & \ddots & \alpha & \beta \end{bmatrix} \quad (2.53)$$

Regardless of whether matrix $\mathbf{B}_2$ from equation 2.50 or matrices $\mathbf{B}_2^{(1)}$ and $\mathbf{B}_2^{(2)}$ from equation 2.53 were used, the resulting solutions to the PDE were virtually identical. Using similar algorithms to those described in section 2.2, graphical solutions were produced (shown in Figure 2.3), and the critical length was determined both visually ([4]) and through computational methods (as shown in Figure 2.4). The computation method approximated the critical length to be 1.579 to three decimal places. This result is consistent with the analytical analysis (for $C = D = 1$) presented in equation 2.46.

2.4 NEUMANN BOUNDARY CONDITIONS

In this section, both boundary conditions for the neutron diffusion equation (2.1) were set to Neumann boundary conditions (2.40 and 2.41). The subsequent analysis will focus on finding the critical length using an analytical approach and a numerical one. From equations 2.7 and 2.8, a solution is obtained in terms of arbitrary constants. Applying the change of variables $y = x + \frac{L}{2}$, and applying the new boundary conditions, the following equations were found:

$$\frac{\partial n(y=0, t)}{\partial y} = Ck \cos(0) - Bk \sin(0) = 0 \quad (2.54)$$

$$\text{therefore } C = 0 \quad (2.55)$$

$$\sin(kL) = 0 \quad (2.56)$$

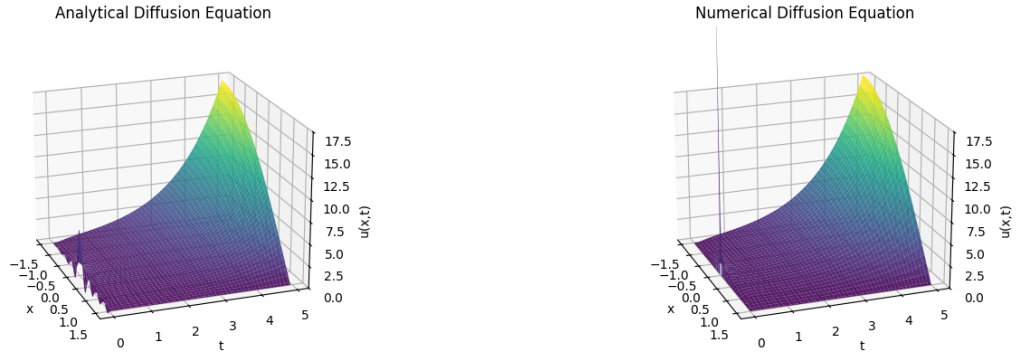
$$k = \frac{n\pi}{L} \quad (2.57)$$

this yields a spatial function of the form:

$$X(x) = B \cos\left(\frac{n\pi}{L}\left(x + \frac{L}{2}\right)\right) \quad (2.58)$$

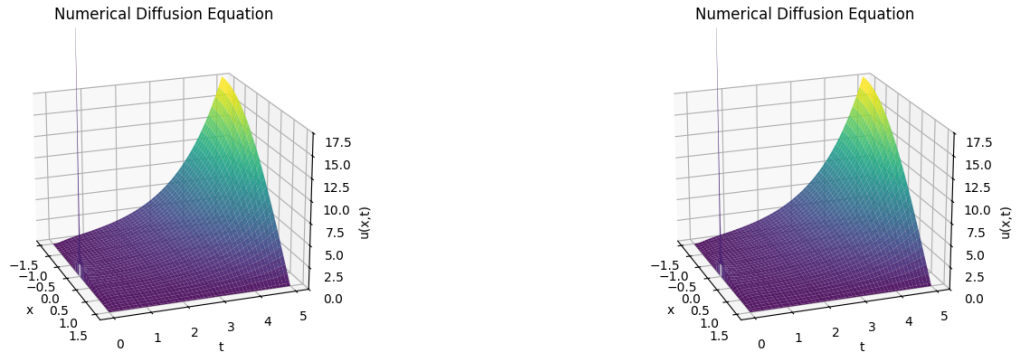
which leads to a solution in terms of the coefficient A_n :

$$n(x, t) = \sum_{n=0}^{\infty} A_n \exp\left(\left(C - \frac{Dn^2\pi^2}{L^2}\right)t\right) \cos\left(\frac{n\pi}{L}\left(x + \frac{L}{2}\right)\right) \quad (2.59)$$



(a) Analytical solution to the neutron diffusion PDE (2.1), using equation 2.47 for $m = 0, 1, \dots, 15$.

(b) Numerical solution to the neutron diffusion PDE (2.1), using the coefficient matrix from equation 2.50.



(c) Numerical solution to the neutron diffusion PDE (2.1), using the coefficient matrix $B_2^{(1)}$ from equation 2.53.

(d) Numerical solution to the neutron diffusion PDE (2.1), using the coefficient matrix $B_2^{(2)}$ from equation 2.53.

Figure 2.3: Plots (a), (b), (c), and (d) present an analytical and multiple numerical solutions to equation 2.1 for boundary conditions 2.40 and $n(x = \frac{L}{2}, t) = 0$ from equation 2.2. All graphs have $L = 3$ and have time up to $t = 5$. In each graph, $u(x, t) \equiv n(x, t)$.

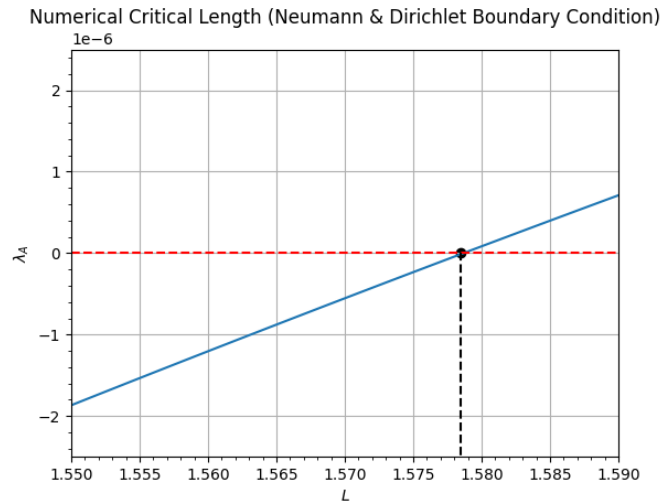


Figure 2.4: An approximation for the critical length was found by an exponential fit described in Section 2.2. λ_A is the gradient to the approximating exponential curve (2.39). The red dashed lined is a visual indication for the x axis, or in this case the λ_A axis. The dashed black line indicates where the approximate critical length $L_{critical}$ is. From the figure, a value for the critical length is:

$$L_{critical} = 1.579 \text{ to three decimal places.}$$

the k value in equation 2.57 has the same form as the k value in section 2.1. However, the key difference here is that the spatial function in equation 2.59 involves a cosine function rather than a sine function (2.17), this means that n can take the value of 0 unlike in the previous case. The solution given in equation 2.59 shows a monotonic increase as a function of time when the exponential term is non-negative. By choosing $n = 0$, the dependence on L is eliminated and the solution grows when C satisfies the following condition:

$$C \geq 0. \quad (2.60)$$

Hence, the solution exponentially increases $\forall L$ if $C \geq 0$. Furthermore, the complete solution to the PDE (2.1) for Neumann boundary conditions (2.40 and 2.41) using the analytical initial condition (2.19) on equation 2.59 was found to be:

$$n(x, t) = \frac{1}{L} e^{Ct} + \frac{2}{L} \sum_{n=1}^{\infty} \cos\left(\frac{n\pi}{2}\right) \exp\left(\left(C - D\left(\frac{n\pi}{L}\right)^2\right)t\right) \cos\left(\frac{n\pi}{L}\left(x + \frac{L}{2}\right)\right) \quad (2.61)$$

$$\text{where } C = D = 1.$$

A numerical approach started from augmenting the coefficient matrix $\mathbf{B}_2^{(1)}$ from equation 2.53, the last row was changed to fit the second Neumann boundary condition 2.41 and a column of zeros was not appended to the solution matrix. Performing a Taylor series of $n(x_{i-1}, t_j)$ around $n(x_i, t_j)$ and truncating $O((\Delta x)^2)$, the two-point backward difference approximation to a first order partial derivative was found:

$$\frac{\partial}{\partial x} n(x_i, t_j) = \frac{n_{i,j} - n_{i-1,j}}{\Delta x} \quad (2.62)$$

applying boundary condition 2.41 to equation 2.62 the following expression was found:

$$n_{N-2,j} = n_{N-1,j} \quad (2.63)$$

this led to the coefficient matrix:

$$\mathbf{B}_3 = \begin{bmatrix} (\beta + \alpha) & 0 & \alpha & & & & \\ & \alpha & \beta & \alpha & & & \\ & & \alpha & \beta & \alpha & & \\ & & & \alpha & \ddots & \ddots & \\ & & & & \ddots & \ddots & \alpha \\ & & & & & \alpha & \beta & \alpha \\ & & & & & \alpha & 0 & (\beta + \alpha) \end{bmatrix} \quad (2.64)$$

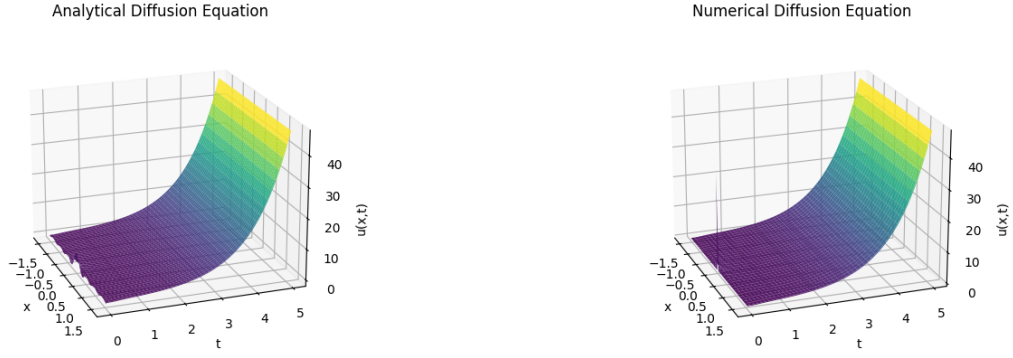
by applying the same algorithm described in section 2.2 solution plots were computed using the coefficient matrix 2.64, the analytical solution 2.61 was also plotted, both graphs are shown in Figure 2.5. As expected, the general form of both the analytical and numerical solutions shown in Figure 2.5 are the same, apart from the beginning where the initial conditions were not equal.

2.5 SURROUNDING THE SYSTEM WITH A TAMPER

In this section, the original Dirichlet boundary conditions were used (2.2) and the spatial domain was scaled by a factor of γ , which led to the possible values for x :

$$-\gamma \frac{L}{2} \leq x \leq \gamma \frac{L}{2}. \quad (2.65)$$

However, C was set equal to zero for $|x| > \frac{L}{2}$. The systems governed by this regime for $\gamma = 2, 6, 12$ were analysed in detail in the subsequent analysis. First, a function in Python was created to retrieve the index of the spatial domain where the first spatial value was greater than $-\frac{L}{2}$, this index is denoted



(a) Analytical solution to the neutron diffusion PDE (2.1), using equation 2.61 for $n = 1, 2, \dots, 15$.

(b) Numerical solution to the neutron diffusion PDE (2.1) with the coefficient matrix 2.64, using the finite difference method discussed in section 2.2.

Figure 2.5: Plots (a) and (b) present an analytical and numerical solution to equation 2.1 for Neumann boundary conditions (2.40 and 2.41) using $L = \pi$ up to time $t = 5$. In each graph, $u(x, t) \equiv n(x, t)$.

Critical Length Approximations

γ	Critical Length
2	1.72
6	0.86
12	0.57

Table 2.1: Critical lengths for varying γ , where γ scales the spatial domain (2.65). Each critical length was approximated using the algorithm discussed in section 2.2, and is given to two decimal places.

q , and the last spatial index corresponding to the last spatial value that was less than $\frac{L}{2}$, this index will be denoted p . An example of the resulting (6×6) coefficient matrix:

$$\mathbf{B}_4 = \begin{bmatrix} \beta_0 & \alpha & & & & \\ \alpha & \beta_0 & \alpha & & & \\ & \alpha & \beta & \alpha & & \\ & & \alpha & \beta & \alpha & \\ & & & \alpha & \beta_0 & \alpha \\ & & & & \alpha & \beta_0 \end{bmatrix} \quad (2.66)$$

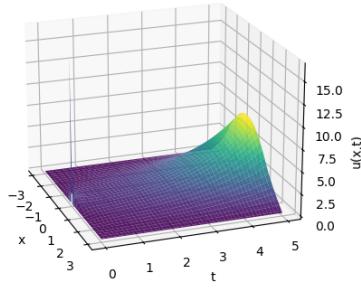
where $\beta_0 = 1 - 2\alpha$, and similar to equation 2.30, $\beta = 1 + C\Delta t - 2\alpha$. The matrix in equation 2.66 has a different diagonal from equation 2.33, if $x_i \in [-\frac{L}{2}, \frac{L}{2}]$ then the corresponding diagonal element was β , otherwise β_0 . The point q in matrix 2.66 corresponds to 2 and p corresponds to 4.⁶ The general diagonal in this regime (2.66) takes the form:

$$\Gamma = [\beta_0 \quad \beta_0 \quad \dots \quad \beta_0 \quad \underbrace{\beta}_q \quad \beta \quad \dots \quad \underbrace{\beta}_p \quad \beta_0 \quad \beta_0 \quad \dots \quad \beta_0] \quad (2.67)$$

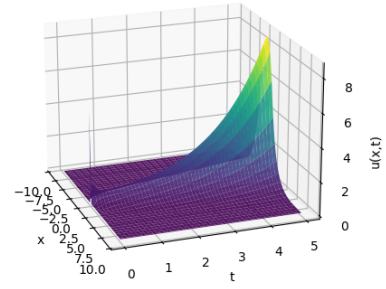
where Γ is a generalised diagonal of the example coefficient matrix 2.66. The resulting solutions using a generalised form of equation 2.66 were plotted and shown in Figure 2.6. The critical length was calculated for $\gamma = 2, 6, 12$ using the same algorithm discussed in section 2.2, the values approximated are shown in Table 2.1 and the graphical representation is shown in Figure 2.7. Finally the critical lengths were visually estimated through varying L and plotting multiple solutions, this is available on GitHub [5].

⁶Using zero-indexing.

Numerical Diffusion Equation $\gamma = 2$

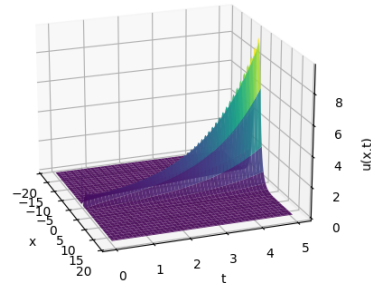


Numerical Diffusion Equation $\gamma = 6$



- (a) A numerical solution to the neutron diffusion PDE (2.1) with Dirichlet boundary conditions (2.2), where the spatial domain has been scaled by $\gamma = 2$ in reference to equation 2.65. (b) A numerical solution to the neutron diffusion PDE (2.1) with Dirichlet boundary conditions (2.2), where the spatial domain has been scaled by $\gamma = 6$ in reference to equation 2.65.

Numerical Diffusion Equation $\gamma = 12$



- (c) A numerical solution to the neutron diffusion PDE (2.1) with Dirichlet boundary conditions (2.2), where the spatial domain has been scaled by $\gamma = 12$ in reference to equation 2.65.

Figure 2.6: Plots (a), (b), and (c) present numerical solutions to equation 2.1 using an adapted version of the example coefficient matrix 2.66, each graph took $L = 3$ and went up to time $t = 5$. In each graph, $u(x, t) \equiv n(x, t)$.

3 WAVE EQUATION

The wave equation of the form:

$$\frac{\partial^2 u}{\partial t^2} = 4 \frac{\partial^2 u}{\partial x^2} \quad (3.1)$$

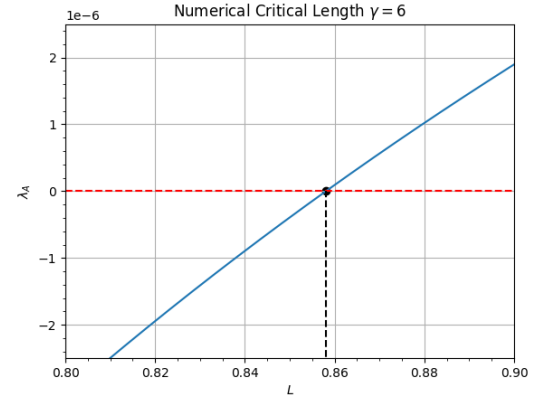
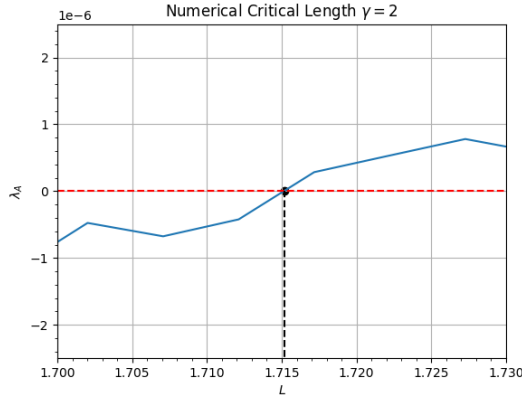
where the speed of light c is equal to two, will be investigated in this paper. First an analytical approach, followed by a finite difference approximation, and finally solving it using the Crank-Nicholson method. The boundary conditions used were Dirichlet conditions, which were:

$$u(x = 0, t) = u(x = 2\pi, t) = 0 \quad (3.2)$$

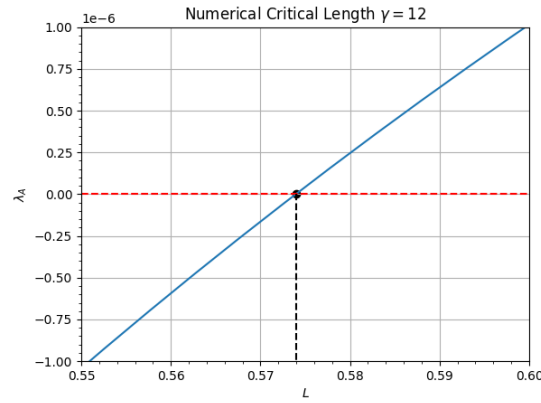
the initial conditions were:

$$u(x, 0) = \begin{cases} 1 & \pi - 1 \leq x \leq \pi + 1, \\ 0 & \text{otherwise.} \end{cases} \quad (3.3)$$

$$\frac{\partial u(x, 0)}{\partial t} = x^2 \sin^2(4x). \quad (3.4)$$



(a) $\gamma = 2$, approximately $L_{critical} = 1.72$ to two decimal places. (b) $\gamma = 6$, approximately $L_{critical} = 0.86$ to two decimal places.



(c) $\gamma = 12$, approximately $L_{critical} = 0.57$ to two decimal places.

Figure 2.7: Plots (a), (b), and (c) present approximations for the critical length determined by the exponential fit described in Section 2.2 for the tamper regime where the spatial domain has been scaled by γ (2.65). λ_A is the gradient to the approximating exponential curve (2.39). The red dashed line is a visual indication for the x axis, or in this case the λ_A axis. The dashed black line indicates where the approximate critical length $L_{critical}$ is. The approximations for each critical length are seen in Table 2.1.

Finally, the allowed spatial and temporal domains were:

$$0 \leq x \leq 2\pi \quad (3.5)$$

$$t \geq 0. \quad (3.6)$$

3.1 ANALYTIC SOLUTION

A solution to equation 3.1 was sought using the method of variable separation, similar to the diffusion equation:

$$\frac{1}{T} \frac{d^2 T}{dt^2} = -\lambda \quad (3.7)$$

$$\frac{4}{X} \frac{d^2 X}{dx^2} = -\lambda \quad (3.8)$$

$$\text{where } u(x, t) = X(x)T(t) \quad (3.9)$$

the solutions to both equation 3.7 and 3.8 are:

$$T(t) = A \cos(k_1 t) + B \sin(k_1 t) \quad \text{where} \quad k_1^2 = \lambda \quad (3.10)$$

$$X(x) = C \cos(k_2 x) + D \sin(k_2 x) \quad \text{where} \quad k_2^2 = \frac{\lambda}{4} \quad (3.11)$$

where A , B , C , and D are all arbitrary constants. An important note regarding equations 3.7 and 3.8 is that, although $\lambda > 0$, a negative sign was assigned to λ in order to obtain solutions to these equations that satisfy the boundary conditions (3.2). Applying boundary conditions 3.2 to equations 3.10 and 3.11, a solution in terms of two unknown coefficients B_n and D_n was found:

$$u(x, t) = \sum_{n=1}^{\infty} \left(B_n \cos(nt) + D_n \sin(nt) \right) \sin\left(\frac{n}{2}x\right). \quad (3.12)$$

The initial condition 3.3 was subsequently applied to equation 3.12, similar to section 2.1, the orthogonality of sinusoidal functions was utilised.

$$\begin{aligned} \int_0^{2\pi} u(x, 0) \sin\left(\frac{m}{2}x\right) dx &= \sum_{n=1}^{\infty} \int_0^{2\pi} B_n \sin\left(\frac{n}{2}x\right) \sin\left(\frac{m}{2}x\right) dx \\ \int_{\pi-1}^{\pi+1} \sin\left(\frac{n}{2}x\right) dx &= \pi B_n \\ B_n &= \frac{4}{n\pi} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n}{2}\right) \end{aligned} \quad (3.13)$$

To find D_n the second initial condition 3.4 was applied to equation 3.12. The methods used to solve for D_n are displayed in the following equations:

$$\begin{aligned} \int_0^{2\pi} u_t(x, 0) \sin\left(\frac{m}{2}x\right) dx &= \sum_{n=1}^{\infty} n D_n \int_0^{2\pi} \sin\left(\frac{n}{2}x\right) \sin\left(\frac{m}{2}x\right) dx \\ D_n &= \frac{1}{n\pi} \int_0^{2\pi} x^2 \sin^2(4x) \sin\left(\frac{n}{2}x\right) dx \end{aligned} \quad (3.14)$$

the integral in equation 3.14 was solved using the method of integration by parts.

$$\int uv' = uv - \int u'v \quad (3.15)$$

$$\text{where} \quad u = x^2 \quad v' = \sin^2(4x) \sin\left(\frac{n}{2}x\right) \quad (3.16)$$

$$u' = 2x \quad v = \int \sin^2(4x) \sin\left(\frac{n}{2}x\right) \quad (3.17)$$

The integral v in equation 3.17 was solved using standard double angle formulas:

$$v = \frac{1}{2} \left(\left(\frac{1}{16+n} \right) \cos\left(\frac{16+n}{2}x\right) + \left(\frac{1}{n-16} \right) \cos\left(\frac{n-16}{2}x\right) - \frac{2}{n} \cos\left(\frac{n}{2}x\right) \right) \quad (3.18)$$

the parts in equation 3.16, 3.17 and 3.18 were combined into equation 3.15, this resulted in a solution for D_n .

$$D_n = \frac{1}{n\pi} \left(2\pi^2 (-1)^n \left(\frac{1}{n+16} + \frac{1}{n-16} - \frac{2}{n} \right) + ((-1)^n - 1) \left(\frac{8}{n^3} - \frac{4}{(n+16)^3} - \frac{4}{(n-16)^3} \right) \right) \quad (3.19)$$

A complete analytical solution to the wave equation (3.1) was found by substituting B_n (3.13) and D_n (3.19) into equation 3.12. The corresponding analytical solution graph is shown in Figure 3.2.

3.2 ANALYTICAL SOLUTION ANALYSIS - STABILITY

The most noticeable aspect in section 3.1 are the singularities found in equation 3.19, where it includes $n - 16$ in the denominator of two terms. The D_n coefficient calculated in equation 3.19 is correct $\forall n \in \mathbb{N} \setminus \{16\}$. However, it fails for $n = 16$, to address this, equation 3.14 was computed for when $n = 16$ and D_n was found to be equal to $-\frac{3\pi}{256}$. To analyse the stability of the analytical solution, the energy of the wave equation was looked at in detail, the energy of the wave equation in one-dimension is given as [6]:

$$E(t) = \frac{1}{2} \int_0^{2\pi} (u_t^2 + 4u_x^2) dx \quad (3.20)$$

using the analytical solution 3.12 obtained:

$$u_t = \sum_{n=1}^{\infty} \left(-nB_n \sin(nt) + nD_n \cos(nt) \right) \sin\left(\frac{n}{2}x\right) \quad (3.21)$$

$$u_x = \sum_{n=1}^{\infty} \frac{n}{2} \left(B_n \cos(nt) + D_n \sin(nt) \right) \cos\left(\frac{n}{2}x\right) \quad (3.22)$$

and substituting these equations (3.21 and 3.22) into the integral in equation 3.20, the following is obtained:

$$\int_0^{2\pi} u_t^2 dx = n^2 \pi (B_n^2 \sin^2(nt) + D_n^2 \cos^2(nt) - 2B_n D_n \sin(nt) \cos(nt)) \quad (3.23)$$

$$4 \int_0^{2\pi} u_x^2 dx = n^2 \pi (B_n^2 \sin^2(nt) + D_n^2 \cos^2(nt) + 2B_n D_n \sin(nt) \cos(nt)) \quad (3.24)$$

$$E = \pi \sum_{n=1}^{\infty} n^2 (B_n^2 + D_n^2). \quad (3.25)$$

Taking the time derivative of equation 3.25 yields zero, since the expression contains no explicit time dependence. This shows total energy is conserved $\forall t$, this implies that the analytical solution is stable as the solution can not grow out of bounds and the total energy never increases as a function of time to become unstable.

The coefficients B_n (3.13) and D_n 3.19 were plotted 3.1 as n was increased to show that the solution (3.12) does not diverge to infinity as $n \rightarrow \infty$. In fact the coefficients B_n and D_n in Figure 3.1 converge to zero. The decay in B_n takes much longer D_n , although by taking the limit as $n \rightarrow \infty$ it is easily shown that:

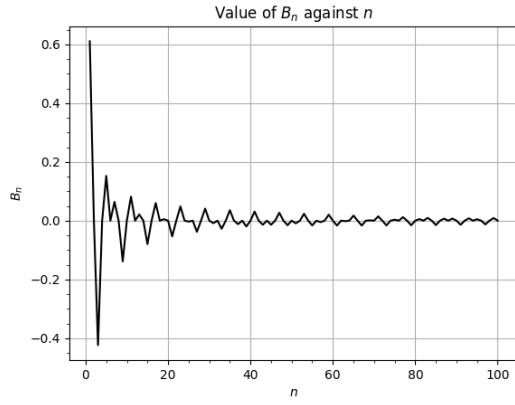
$$\lim_{n \rightarrow \infty} \left(\frac{4}{n\pi} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n}{2}\right) \right) = 0. \quad (3.26)$$

The limit in equation 3.26 along with Figure 3.1 show that the analytical solution to the wave equation is finite and doesn't diverge or blow up.

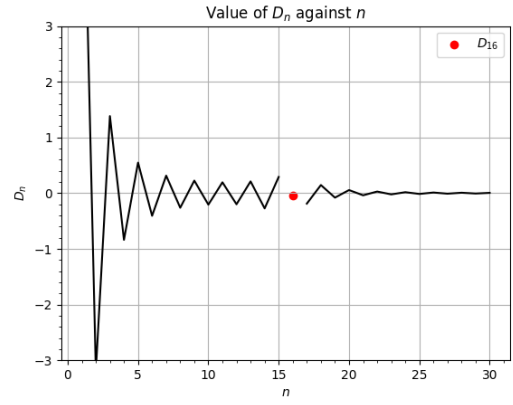
Finally, the initial condition was slightly modified to demonstrate that small changes lead to corresponding small adjustments in the analytical solution, which reflect the perturbations. The new initial condition was:

$$u(x, 0) = \begin{cases} 1 + 0.05 & \pi - 1 \leq x \leq \pi + 1, \\ 0 & \text{otherwise.} \end{cases} \quad (3.27)$$

The wave equation (3.1) with the initial condition 3.27 instead of 3.3 and can be seen in Figure 3.2. In the figure, both systems share the overall same shape and are nearly identical, though a slight difference appears near the peak (yellow in the figure). Two small dips create a visible line, which is a result of the perturbation. Despite this, the strong agreement between the unperturbed and perturbed systems demonstrates that the analytical solution exhibits good stability.

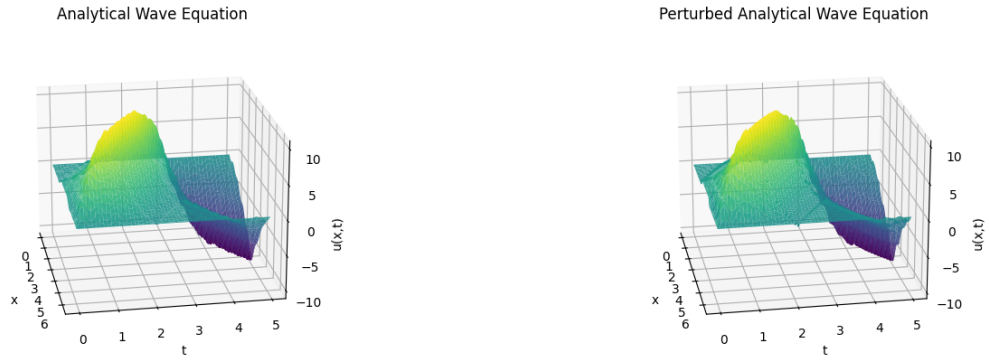


(a) The value of B_n from equation 3.13, for values of n in steps: 1, 2, ..., 100.



(b) The value of D_n from equation 3.19, for values of n in steps: 1, 2, ..., 30.

Figure 3.1: The red dot (D_{16}) in plot (b) shows the singularity that was evaluated with equation 3.14 rather than equation 3.19.



(a) Analytical wave solution using the solution discussed in section 3.1, where $n = 1, 2, \dots, 100$.

(b) Perturbed wave solution using the initial condition 3.27.

Figure 3.2: Plots (a) and (b) present an analytical solutions to equation 3.1.

3.3 NUMERICAL SOLUTION

A grid of values for both the spatial and temporal domains (3.5 and 3.6), similar to the grids shown in equations 2.22 and 2.23, was defined for the wave system.

$$x_i = i\Delta x \quad \text{for } i = 0, 1, 2, \dots, N-1 \quad \text{where} \quad \Delta x = \frac{2\pi}{N-1} \quad (3.28)$$

$$t_j = j\Delta t \quad \text{for } j = 0, 1, 2, \dots \quad (3.29)$$

A forward-in-time finite difference equation for the wave equation (3.1) was derived by approximating the derivatives (u_{tt} and u_{xx}) by the three-point central difference approximations

$$u_{i,j+1} = \alpha u_{i-1,j} + \beta u_{i,j} + \alpha u_{i+1,j} - u_{i,j-1} \quad (3.30)$$

$$\text{where } \alpha = \frac{4(\Delta t)^2}{(\Delta x)^2} \quad \beta = 2(1 - \alpha). \quad (3.31)$$

Equation 3.30 has a similar form to the finite difference equation found in section 2.2 equation 2.29. However, an additional term involving a backward time step now appears. Equation 3.30 can be

expressed in matrix form in the following way:

$$\mathbf{U}^{(j+1)} = \mathbf{A}\mathbf{U}^{(j)} - \mathbf{U}^{(j-1)} \quad \text{where} \quad \mathbf{U}^{(j)} = \begin{bmatrix} u_{1,j} \\ u_{2,j} \\ \vdots \\ u_{N-2,j} \end{bmatrix} \quad (3.32)$$

where A takes the same form as the matrix in equation 2.33 and represents the coefficients for the interior values, that exclude the boundary conditions. By applying the initial conditions to the system, they produce the column vectors $\mathbf{U}^0$ and $\mathbf{U}^1$, starting with initial condition 3.3, this provides a column vector in the form:

$$\mathbf{U}^{j=0} = \begin{bmatrix} u_{1,0} \\ u_{2,0} \\ \vdots \\ u_{N-2,0} \end{bmatrix} = [0 \quad 0 \quad \dots \quad 0 \quad 1 \quad \dots \quad 1 \quad 0 \quad \dots \quad 0]^T \quad (3.33)$$

where the solutions $u_{x,0}$ are equal to one when $x \in [\pi - 1, \pi + 1]$, and they are equal to zero $\forall$ other values of x . Applying the initial condition 3.4 and employing the two-point forward approximation, the components that constitute $\mathbf{U}^1$ were computed:

$$u_{i,1} = \begin{cases} \Delta t x_i^2 \sin^2(4x_i) & \text{for } x_i \notin [\pi - 1, \pi + 1] \\ \Delta t x_i^2 \sin^2(4x_i) + u_{i,0} & \text{for } x_i \in [\pi - 1, \pi + 1] \end{cases} \quad (3.34)$$

using the same algorithm outlined in section 2.2 for solving U^{j+1} , the solution to equation 3.1 was computed using equation 3.32, alongside the initial column vectors 3.33 and 3.34. The resulting solution is shown in Figure 3.3.

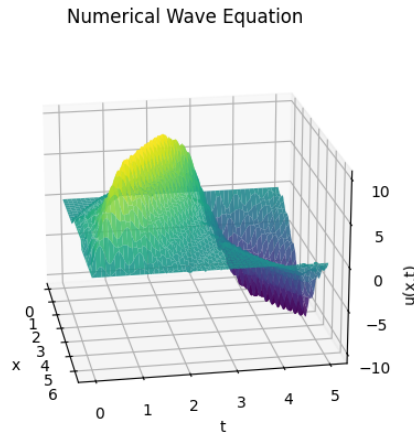


Figure 3.3: The numerical solution to the wave equation 3.1 using boundary condition 3.2 and initial conditions 3.3 and 3.4, using equation 3.32.

3.4 ANALYTICAL AND NUMERICAL COMPARISON

To compare the analytical and numerical solutions to the wave equation 3.1, described in sections 3.1 and 3.3 respectively, the grid sizes in both the temporal and spatial domains were made equal.⁷ The exact size of the numerical and analytical solution matrices were: 10000×101 . The relative Euclidean norm was computed between the solutions to measure the distance between the numerical and

⁷ $\Delta t \neq \Delta x$. Δt for the numerical grid was the same as the analytical Δt defined in Python, similar with Δx .

analytical results. A smaller relative Euclidean norm indicates a strong agreement between the solutions. The relative Euclidean norm, denoted as $\|e\|_{2,rel}$, was given by the following expressions [7]:

$$\|e\|_2 = \sqrt{\sum_{i,j} |u_N(x_i, t_j) - u_A(x_i, t_j)|^2 \Delta x \Delta t} \quad (3.35)$$

$$\|u\|_2 = \sqrt{\sum_{i,j} |u_A(x_i, t_j)|^2 \Delta x \Delta t} \quad (3.36)$$

$$\|e\|_{2,rel} = \frac{\|e\|_2}{\|u\|_2} \quad (3.37)$$

where u_N is the numerical solution to the wave equation and u_A is the analytical solution. The value computed for the relative Euclidean norm was: 0.0364 to three decimal places. As stated previously, since the calculated value of $\|e\|_{2,rel}$ is small, this suggests strong agreement between the numerical and analytical solutions.

3.5 CRANK-NICHOLSON

To acquire a more stable finite difference approximation to the wave equation (3.1), the Crank-Nicholson (CN) method was employed. The CN forward time step approximation to the wave equation 3.1 was found to be [8]:

$$\frac{\alpha}{2} u_{i+1,j+1} - (\alpha + 1) u_{i,j+1} + \frac{\alpha}{2} u_{i-1,j+1} = -\frac{\alpha}{2} u_{i+1,j} + (\alpha - 2) u_{i,j} - \frac{\alpha}{2} u_{i-1,j} + u_{i,j-1} \quad (3.38)$$

$$\text{where } \alpha = \frac{4(\Delta t)^2}{(\Delta x)^2} \quad (3.39)$$

the corresponding matrix form of equation 3.38 was:

$$\mathbf{P}\mathbf{C}^{j+1} = \mathbf{Q}\mathbf{C}^j - \mathbf{C}^{j-1} \quad (3.40)$$

where $\mathbf{C}^j$ is the j time step solution column vector, similar to that seen in equation 3.32. Finally, $\mathbf{P}$ and $\mathbf{Q}$ are the inner tridiagonal coefficient matrices with the boundary conditions applied 3.2. An additional note on equation 3.40, taking the normal inverse of matrix $\mathbf{P}$ to solve for $\mathbf{C}^{j+1}$ was not done, instead the equation was solved using the Thomas Algorithm [9] for $\mathbf{C}^{j+1}$, this was because $\mathbf{P}$ was ill-conditioned.

$$\mathbf{P} = \begin{bmatrix} \eta & \zeta & & & \\ \zeta & \eta & \zeta & & \\ & \zeta & \eta & \zeta & \\ & & \zeta & \ddots & \ddots \\ & & & \ddots & \ddots & \zeta \\ & & & & \zeta & \eta \end{bmatrix} \quad \mathbf{Q} = \begin{bmatrix} \phi & -\zeta & & & \\ -\zeta & \phi & -\zeta & & \\ & -\zeta & \phi & -\zeta & \\ & & -\zeta & \ddots & \ddots \\ & & & \ddots & \ddots & -\zeta \\ & & & & -\zeta & \phi \end{bmatrix} \quad (3.41)$$

$$\text{where } \eta = -(\alpha + 1), \quad \zeta = \frac{\alpha}{2}, \quad \text{and } \phi = (\alpha - 2). \quad (3.42)$$

Applying the initial condition 3.3 resulted in the same initial column vector seen in equation 3.33. However, the second solution column vector was found by performing a Taylor expansion of $u_{i,1}$ around $u_{i,0}$ and replacing the second time derivative with $4u_{xx}$ and substituting in the central difference approximation for u_{xx} . Applying the second initial condition (3.4) yielded the following constituent elements that make up the second solution vector:

$$u_{i,1} = u_{i,0} + \Delta t x_i^2 \sin^2(4x_i) + \frac{\alpha}{2} (u_{i-1,0} - 2u_{i,0} + u_{i+1,0}) \quad (3.43)$$

$$\text{for } u_{i,0} = \begin{cases} 1 & \text{if } x_i \in [\pi - 1, \pi + 1] \\ 0 & \text{if } x_i \notin [\pi - 1, \pi + 1] \end{cases} \quad (3.44)$$

the algorithm discussed in section 2.2 to calculate matrix equations into PDE solutions was used and the following solution was plotted in Figure 3.4.

Analysing both the explicit central finite difference performed in equation 3.30 and CN (3.38) via von Neumann analysis, the condition for stability for the central finite difference was found to be:

$$\frac{1}{2} \geq \frac{\Delta t}{\Delta x} \quad (3.45)$$

using the same analysis, the Crank-Nicholson method was proven to be unconditional, resulting in a stable solution for all Δt and Δx . Using the same grid size, the CN method resulted in a relative Euclidean norm (3.37) of 0.0342, which is more accurate than the central difference method. However, for a similar relative Euclidean norm, as long as the stability condition 3.45 is met the explicit finite difference method is extremely effective and efficient.

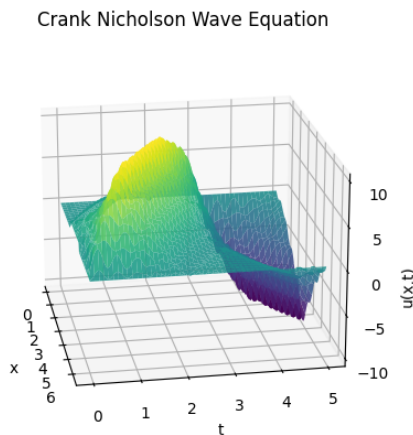


Figure 3.4: The solution via Crank-Nicholson method to the wave equation 3.1 using boundary condition 3.2 and initial conditions 3.3 and 3.44, using equation 3.40.

REFERENCES

- [1] J. D. Hunter, "Matplotlib: A 2d graphics environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.
- [2] L. A. Miller, "PDE solution animations," 2025. [Online]. Available: https://github.com/louisaaronmiller/PDEproject/tree/main/Gifs_question_1_part_A
- [3] F. L. Floc'h, "Instabilities of explicit finite difference schemes with ghost points on the diffusion equation," 2023. [Online]. Available: <https://arxiv.org/abs/2308.04629>
- [4] L. A. Miller, "PDE solution animations," 2025. [Online]. Available: https://github.com/louisaaronmiller/PDEproject/tree/main/Gifs_question_1_part_B
- [5] L. A. Miller, "PDE solution animations," 2025. [Online]. Available: https://github.com/louisaaronmiller/PDEproject/tree/main/Gifs_question_1_part_D
- [6] V. Grigoryan, "Math 124a lecture 7: The energy method," October 2011, lecture notes.
- [7] S. González-Pinto, E. Hairer, and D. Hernández-Abreu, "Pde-convergence in euclidean norm of amf-w methods for multidimensional linear parabolic problems," *ESAIM: Mathematical Modelling and Numerical Analysis*, vol. 58, pp. 2119–2133, 2024.
- [8] J. Degefa, "Numerical solution of two-dimensional wave equation using crank–nicolson scheme," M.Sc. Thesis, Jimma University, Jimma, Ethiopia, 2018.
- [9] N. J. Higham, *Accuracy and Stability of Numerical Algorithms: Second Edition*. SIAM, 2002.